MADHAV SINGH BHANOT

AI Full-Stack Engineer | React, TypeScript, Node.js, Python, LLM Systems

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SUMMARY

Full-stack engineer (B.Tech CSE, June 2026) who built LLM research tooling end to end at a Helsinki AI startup: the agent and artifact UI in React + TypeScript, real-time WebSocket and webhook backends in Node.js, RBAC, PostgreSQL, and Airflow/Celery ingestion and LLM evaluation pipelines in Python. Before that, built a full Instagram-style social app (React Native, Node.js, MongoDB, Socket.IO, S3) at a Bangalore startup. Open-source contributor to uptime-kuma (92k+ GitHub stars).

SKILLS

- **Frontend:** React 19, TypeScript, Next.js (SSR/SSG), React Native (Expo), Vue.js, Tailwind CSS, Framer Motion, IndexedDB
- **Backend:** Node.js, Express.js, REST APIs, WebSockets, Socket.IO, Webhooks, Pub/Sub, RBAC, OAuth 2.0, JWT, Prisma, Mongoose
- **AI & Data:** OpenAI, Anthropic & Gemini APIs, RAG, LLM-as-Judge Evals, LangSmith, Pydantic, Apache Airflow, Celery, Parquet
- **Databases & Cloud:** PostgreSQL (migrations), MongoDB, Vector DBs, Docker, AWS (EC2, S3 presigned uploads), Vercel
- **Languages & Tools:** JavaScript (ES6+), TypeScript, Python, Java, SQL, Jest, Git, Postman, Jira

EXPERIENCE

Innate AI | Full-Stack / React Developer (Internship) | Helsinki, Finland (Remote) | Jul 2025 - Oct 2025

- Built the React + TypeScript interface of an LLM research platform used by VC advisory teams: step-wise agent “thinking” view, live sources panel, and Claude-style editable report artifacts on a custom Markdown editor and parser.
- Built a real-time **Signals** pipeline: Airflow DAGs detect new articles and media, fire a Node.js webhook, and the backend pushes notifications and fresh data to React over WebSockets (pub/sub); articles open in-app via proxied iframes.
- Wrote the Node.js/Express WebSocket layer for concurrent multi-chat rooms, streaming live LLM execution state to the UI instead of HTTP polling (~300ms faster updates).
- Implemented multi-tenant **RBAC over OAuth** across API middleware and frontend protected routes; wrote PostgreSQL migrations (Postgres and Airflow run in Docker) and stored S3 media references in Postgres.
- Built file parsers (PDF, DOCX, HTML, more) and the Airflow DAG running them on autoscaling Celery workers (1,000+ jobs per worker) to feed a RAG ingestion pipeline (Pydantic, vector DB), exporting JSON and Parquet.
- Built an async **LLM-as-judge eval harness** (GPT-4o-mini, Gemini 2.0 Flash) scoring retrieval, summarization and relevancy, traced in LangSmith.
- Prototyped analysis dashboards with the VC team from PubMed data (SWOT bento grids, competitor analysis, drug-label ingest); fixed out-of-sync global filter stores, shipped company-wide dark mode and regex-highlighted artifact previews.

Krishlabs | Junior Full-Stack Developer (Contract) | Bangalore, India | Nov 2024 - Apr 2025

- Built an Instagram-style social app end to end (Expo React Native, Node.js/Express, MongoDB, 50+ REST endpoints): reels, posts with in-app photo editing, stories with replies, nested comment threads, likes, shares, saves, follow/block/privacy.
- Built a **personalized feed** ranking posts by interest tags, followed accounts and trending engagement, with a cold-start fallback for new users.
- Built real-time 1:1 and group chat over **Socket.IO** with image, audio and document messages, online presence, and Expo push notifications with unread counts.
- Moved media uploads to direct-to-**S3 presigned URLs**; secured the EC2-hosted API with Clerk + Google/GitHub OAuth, JWT refresh tokens, Joi validation, Helmet, rate limiting and XSS/NoSQL-injection sanitization.
- Built an admin job board (post, apply, tag search) and shipped the Next.js landing page with a MongoDB-backed waitlist.

OPEN SOURCE

- **louislam/uptime-kuma (92k+ stars) - PR #7732, in review (2026):** Vue.js feature syncing monitor filters with URL query params so filtered views persist and are shareable; +336 lines incl. a 206-line test spec.
- **lingdojo/kana-dojo (3.6k+ stars) - PR #28654, merged (2026):** Contributed a new community UI theme, merged by maintainers.

PROJECTS

Gaminai - AI Research Assistant | React, TypeScript, Express.js, PostgreSQL, Prisma | Live Demo

- Full-stack RAG assistant over OpenAI and Anthropic APIs with streamed answers, live web retrieval that cites current sources, and persistent multi-turn memory with context compression.

ProScape-Focus - Offline-First Focus Workspace | React 19, IndexedDB, Express.js, OAuth 2.0 | Live | GitHub

- Offline-first workspace syncing IndexedDB state through a 600ms debounced buffer; Express backend with Google OAuth and a caching layer over the YouTube API that cut API latency by 50%.

RunAway - Interactive Audio & Canvas Studio | Vanilla JS, Canvas 2D, Web Audio API, Node.js | Live | GitHub

- Framework-free drawing app (30+ daily users) with a procedural Web Audio rain synth, zero-dependency Node.js dev server and a 358-test unit suite.

EDUCATION & ACHIEVEMENTS

Bharati Vidyapeeth Deemed University | Pune, India | B.Tech CSE | CGPA 8.32/10 | June 2026

- **Hackathon:** 3rd place out of 200+ onsite participants at the Dev with AI Hackathon (Unstop).